

RAFFLES INSTITUTION
2012 YEAR 6 PRELIMINARY EXAMINATION
Higher 3



PHARMACEUTICAL CHEMISTRY

9812/01

Paper 1

18 September
2 hours 30 minutes

Additional Materials: Answer Paper
 Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your index number, civics tutorial group and name on the Cover Page.

Write in dark blue or black pen on both sides of the writing paper.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer any **five** questions.

Begin each question on a fresh sheet of paper.

At the end of the examination, fasten all your work securely together, with the cover page on top.

The number of marks is given in brackets [] at the end of each question or part question.

You may use a calculator.

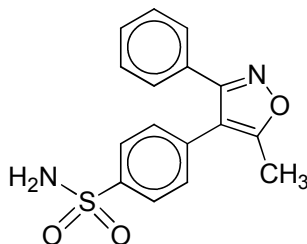
You are reminded of the need for good English and clear presentation in your answers.

This document consists of **13** printed pages and **1** blank page.

[Turn over

- 1 Valdecoxib is a non-steroidal anti-inflammatory drug (NSAID) which was approved in 2001 for the treatment of osteoarthritis, rheumatoid arthritis, and painful menstrual symptoms. However it was removed from the market in 2005 due to concerns about possible increased risk of heart attack and stroke.

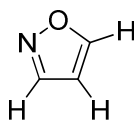
Valdecoxib works as a cyclooxygenase-2 (COX-2) selective inhibitor by competitively binding to a valine residue, $-\text{COCH}(\text{CH}(\text{CH}_3)_2)\text{NH}-$, of the enzyme.



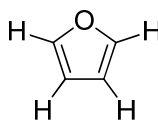
valdecoxib

- (a) (i) Explain the term *competitive inhibition*. [2]
- (ii) By considering the drug-receptor interaction, explain how valdecoxib provides relief from symptoms of inflammation and pain. [2]

Valdecoxib is unique in its design as it contains an aromatic isoxazole ring, an analogue of furan.



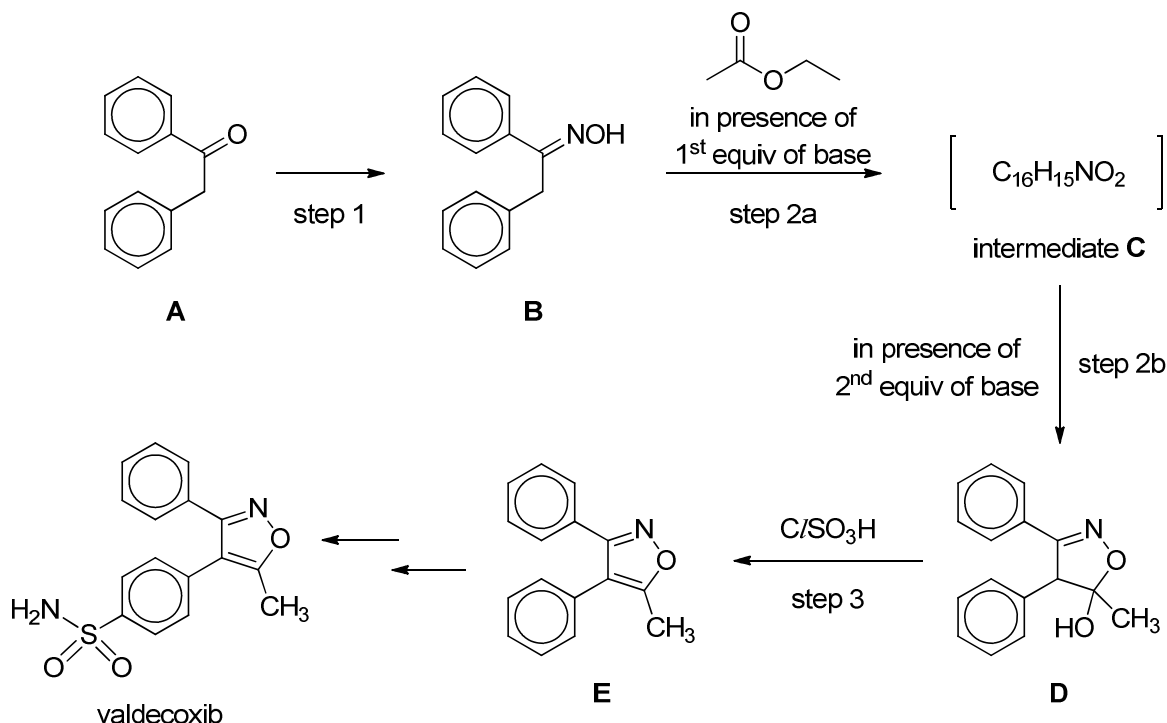
isoxazole



furan

- (b) Deduce whether isoxazole, when compared to furan, is more or less susceptible to electrophilic aromatic substitution. [2]

Valdecoxib can be synthesised in the laboratory through the following scheme which assembled the isoxazole ring elegantly.



(c) (i) Suggest the reagent to be used in step 1. [1]

(ii) Draw the pair of stereoisomers present in **B** and state their configurations. [2]

1 molar equivalent of ethyl ethanoate, CH₃CO₂CH₂CH₃, and 2 molar equivalents of a strong base are needed for the entire step 2. Water is added only at the end of step 2b to give **D**.

(iii) Copy the structure of **B** onto your answer script. On the diagram, label the most acidic proton in **B** with K_{a1} , and the second most acidic proton in **B** with K_{a2} . [1]

(iv) Using your answer in (c)(iii), suggest the identity of intermediate **C**. [1]

(v) What *type of reaction* is step 2b? Suggest a mechanism for this reaction. [3]

(vi) Using an appropriate Newman projection, describe the E2 mechanism for step 3. [2]

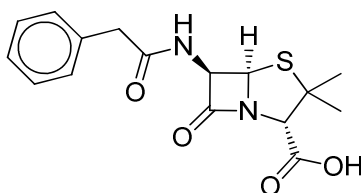
(d) (i) Provide a reason why not all of the vibrational modes of a molecule may be observed in its IR spectrum. [1]

(ii) With reference to the *Data Booklet*, suggest a wavenumber (in cm⁻¹) for the IR stretching frequency of C=N in **B**. Explain briefly. [2]

(iii) With reference to the *Data Booklet*, suggest how **A** can be differentiated from **B** using IR spectroscopy. [1]

[Turn over

- 2 Penicillin G is a β -lactam antibiotic. It interferes with the production of peptidoglycan (an important component of the bacterial cell wall) by binding to bacterial transpeptidases, also known as penicillin-binding proteins (PBPs).



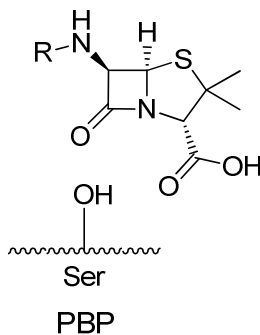
penicillin G

- (a) State the role of bacterial transpeptidase and hence deduce if penicillin G acts as a bactericidal or bacteriostatic drug. [2]

PBP contains a serine residue in its active site which interacts with penicillin, resulting in the irreversible inhibition of the enzyme.

- (b) (i) What is meant by the term *irreversible inhibition* of an enzyme? [1]

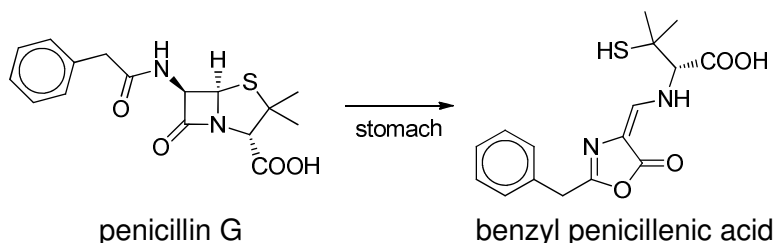
- (ii) Describe the mechanism for the irreversible inhibition of PBP, using a β -lactam antibiotic and the active site of PBP shown below.



[3]

- (iii) Explain why the free carboxylic acid in penicillin G is essential for the drug to function. [1]

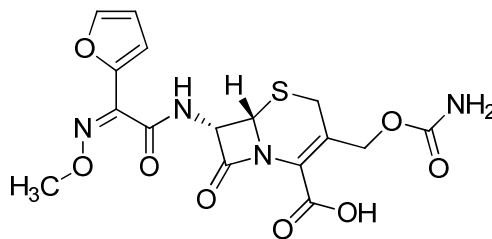
- (c) Penicillin G cannot be administered orally due its susceptibility to acidic hydrolysis in the stomach. Furthermore, the neighbouring acyl group of the amide bond in penicillin G can participate in an intramolecular process to open up the β -lactam ring as illustrated below.



- (i) Propose a mechanism for this self-destructive process, forming benzyl penicillenic acid. [3]

- (ii) Draw the structure of an analogue of penicillin G which you would design to minimise its acid sensitivity. [1]

About 10% of Americans report an allergy to penicillin or a related antibiotic. Clinical trials have shown that structurally-related antibiotics, such as cefuroxime, should be avoided by patients allergic to penicillin.



cefuroxime
(molar mass = 424 g mol^{-1})

- (d) The amount of cefuroxime available in a 500 mg tablet may be estimated by UV spectroscopy. The maximum absorbance occurs at a wavelength of 700 nm with the molar absorptivity at $3.0 \times 10^4 \text{ dm}^3 \text{ mol}^{-1} \text{ cm}^{-1}$.

(i) Explain why cefuroxime is able to absorb UV radiation. [1]

Therapeutic window is a well-defined range of a drug's serum concentration at which a desired effect occurs, below which there is little effect, and above which toxicity occurs.

The therapeutic window for cefuroxime was determined to be between 2 and 8 mg dm^{-3} .

(ii) Calculate the desired average concentration of cefuroxime in mol dm^{-3} of serum. [1]

(iii) Using a path length of 1 cm, calculate the absorbance at the desired average concentration of cefuroxime in serum. [1]

Cefuroxime is taken by a patient twice a day (12-hour interval).

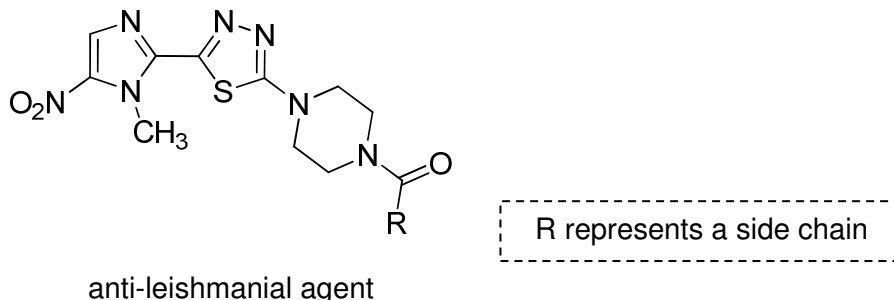
Each dose consists of a single 500 mg tablet.

The half-life of cefuroxime in the body is 1.5 hours.

- (e) (i) Calculate, to three decimal places, the equilibrium maximum and minimum amount (in mg) of cefuroxime in the body. [2]
- (ii) After a few days on the antibiotic, a patient realised that he had missed his 8.00 a.m. dose and decided to take a double dose at 8.00 p.m. on the same day to "compensate" for it. Calculate what the maximum and minimum amount (in mg), to four significant figures, would be for the next 12 hours after taking the double dose. [2]
- (iii) The same non-compliant patient decided to stop his course of antibiotic as the symptoms of his infection have disappeared. Explain why he should have finished the entire course. [2]

- 3 Leishmaniasis is a spectrum of diseases caused by intracellular protozoan parasites and is transmitted by the bite of a certain species of sand fly. Common symptoms include skin sores, prolonged fever, weight loss as well as anaemia. The World Health Organisation has identified leishmaniasis as a major public health problem.

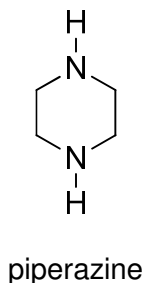
The use of nitroimidazole derivatives as anti-protozoal agents is well established. The general structure of an active anti-leishmanial agent is shown below.



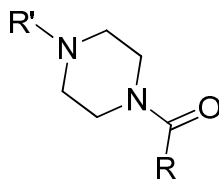
Anti-protozoal agents work similarly to antibiotics. Anti-leishmanial agents typically function by inhibiting protein synthesis.

- (a) Describe **three other ways** in which antibiotics act. [3]

An anti-leishmanial agent contains a 6-membered piperazine ring which can exist in either the chair or boat conformation.

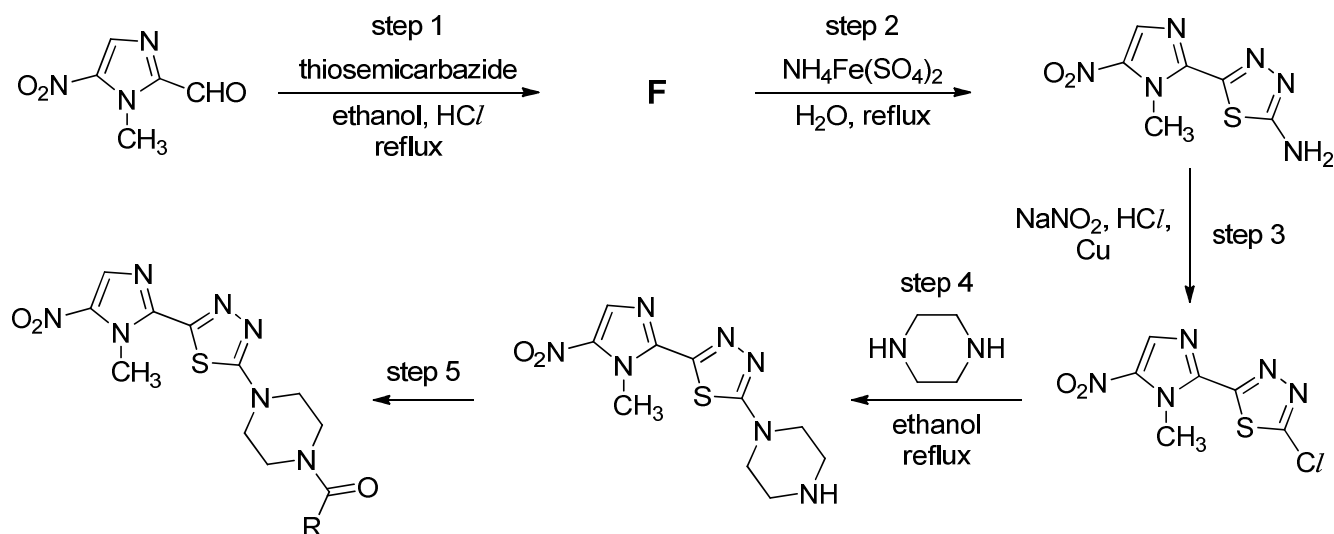


- (b) Draw the **most unstable boat conformation** and the **most stable chair conformation** for the 6-membered piperazine ring in an anti-leishmanial agent. Explain your answer briefly. You may simplify the structure of an anti-leishmanial agent as below.

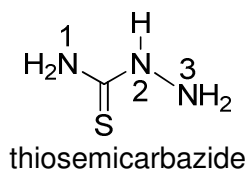


[3]

The synthesis of an anti-leishmanial agent is shown below.



(c) (i) The structure of thiosemicarbazide used in step 1 is shown below.



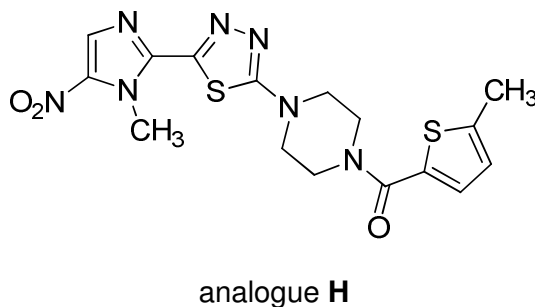
Explain, with the aid of appropriate diagrams, why N-3 is the **most nucleophilic** of the three N atoms in thiosemicarbazide. [2]

(ii) Using the information in (c)(i), state the *type of reaction* for step 1 and suggest the mechanism. Show clearly the structure for **F**. [5]

(iii) What *type of reaction* is step 4? Suggest a mechanism for this reaction. [3]

(iv) Suggest the reagent and condition for step 5. [1]

In 2009, Poorrajab *et al.* discovered the following analogue **H** of an anti-leishmanial agent that displayed the highest activity with the receptor of the protozoa *Leishmania major* which is responsible for Leishmaniasis.

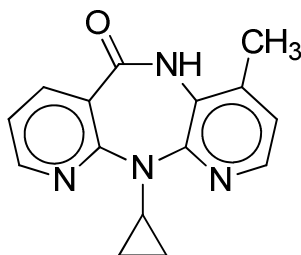


(d) (i) Suggest the **two** types of drug-receptor interactions that might be present between **H** and the receptor site. Indicate the parts of **H** that can participate in such interactions. [2]

(ii) Suggest why **H** is suitable to be administered orally. [1]

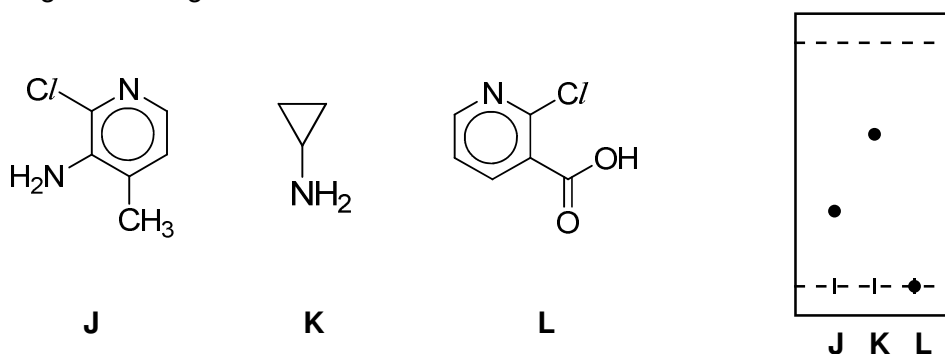
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4 Marketed under the trade name Viramune®, nevirapine is used widely to combat the HIV-1 infection.



nevirapine

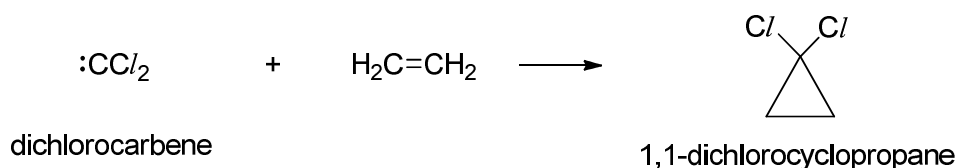
- (a) The synthesis of nevirapine in the laboratory was achieved using **J**, **K** and **L** as starting materials. A TLC analysis of these three compounds was carried out on silica gel using an eluent mixture of dichloromethane and methanol, added with a few drops of aqueous ammonia. The resulting chromatogram is shown below.



- (i) Which of the three compounds is the **most** polar in this system? Explain briefly. [1]
- (ii) Explain the relative positions of **J** and **K**. [1]
- (iii) Sketch the TLC chromatogram if the same eluent mixture is now added with a few drops of dilute hydrochloric acid instead of aqueous ammonia. [1]

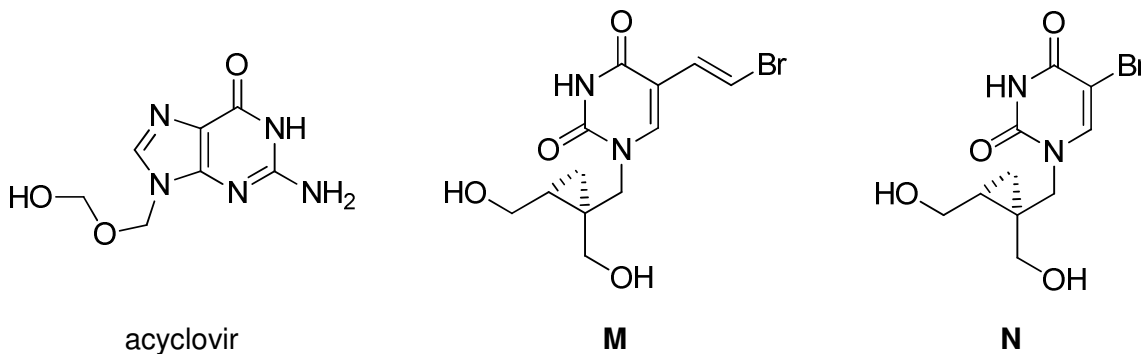
A key feature in the drug design of nevirapine lies in the cyclopropane ring. Its compact and rigid conformation allows optimal fit in tight hydrophobic pockets of many enzymes.

- (b) Cyclopropane can be synthesised from alkene and dichlorocarbene, which contains a lone pair of electrons on the carbon atom.



- (i) State the **two** types of strains present in 1,1-dichlorocyclopropane. [1]
- (ii) With the aid of a suitable diagram, explain why the carbon in :CCl_2 is **both** nucleophilic and electrophilic at the same time. [2]
- (iii) Using your diagram in (b)(ii) and the information above, outline the mechanism for the **concerted** reaction between :CCl_2 and *trans*-but-2-ene to give the (*R,R*) and (*S,S*) racemates. Assign all chiral centres of your products as *R* or *S*. [3]

In 2000, Tsuji and co-workers designed compounds **M** and **N** to include the cyclopropane moiety as analogues of acyclovir – a common anti-viral drug for chickenpox. Notably, **M** was found to be 40-60 times more potent than acyclovir against clinical isolates of the anti-varicella zoster virus but **N** had limited inhibitory activity.



- (c) The syntheses of **M** and **N** began with a chiral compound **W** which contains a cyclopropane ring. **W** consists of only carbon, hydrogen and oxygen atoms. The mass spectrum of **W** shows a molecular ion at m/e 170. The intensity of the m/e 171 peak is 8.9% that of the molecular ion.

The ^1H NMR spectral data of **W** in the presence of CDCl_3 is summarised below.

δ / ppm	integration	splitting
0.88	2	doublet
1.23	3	triplet
2.07	1	multiplet
4.21	2	quartet
4.26	2	doublet

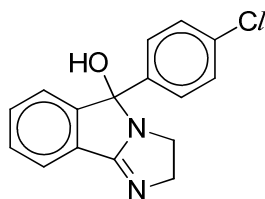
None of these signals disappears when D_2O is used instead of CDCl_3 .

The 1,4-diol moiety in **M** and **N** is achieved by treating **W** with LiAlH_4 .

- (i) Outline the principles of ^1H NMR Spectroscopy. [2]
- (ii) Use the data to work out the full structure of **W**, ignoring all stereochemical configurations. Explain the logic behind your deductions. [6]
- (iii) When Tsuji uses **W** to synthesise either **M** or **N**, the stereochemistry of **W** was retained. Draw another structure of **W**, showing its appropriate stereochemistry with bold wedged or hashed wedged bonds. [1]
- (iv) **M** and **N** are analysed by UV spectroscopy. Predict and explain briefly which compound will have its maximum absorption peak at a shorter wavelength. [2]

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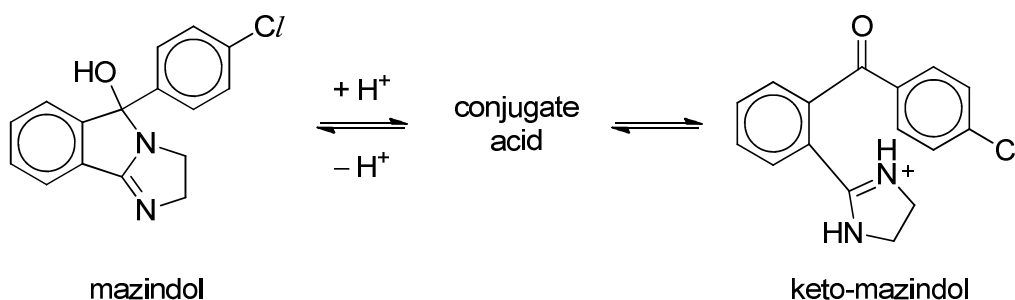
- 5 Over the years, many cocaine analogues have been designed by modifying the chemical structure of cocaine in ways which retain its useful stimulant or antidepressant pharmacological effects, minimise its high toxicity and dependence liability, and yield a competitive cocaine antagonist. Mazindol is a cocaine antagonist which is not euphorigenic.



mazindol

- (a) (i) What do you understand by the term *stimulant*? [1]
- (ii) Explain the difference between the terms *agonist* and *antagonist*. [2]

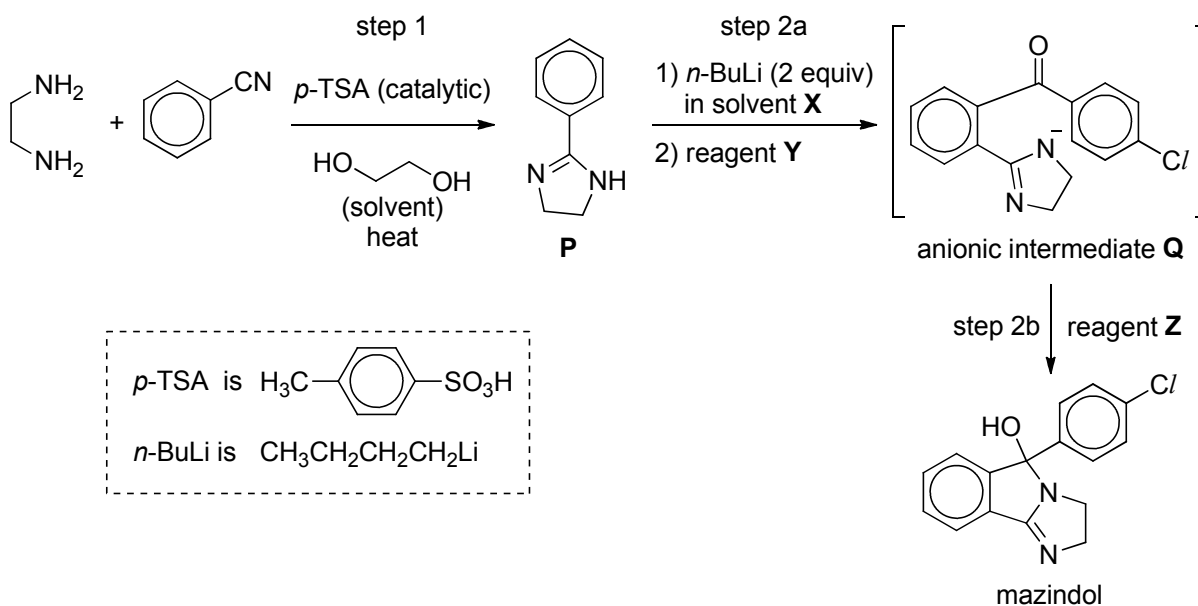
In aqueous acidic medium, mazindol is known to establish the following behaviour.



Mazindol inhibits dopamine and serotonin re-uptake by binding to the dopamine active transporter (DAT) protein with its keto form.

- (b) (i) Draw the resonance structures of the conjugate acid of mazindol. [1]
- (ii) Propose a mechanism for the conversion of the conjugate acid to keto-mazindol. [1]
- (iii) Explain which compound, mazindol or keto-mazindol, is better able to pass through the blood-brain barrier. [2]

The synthesis of mazindol is shown below:



(c) (i) Suggest the mechanism for step 1. [3]

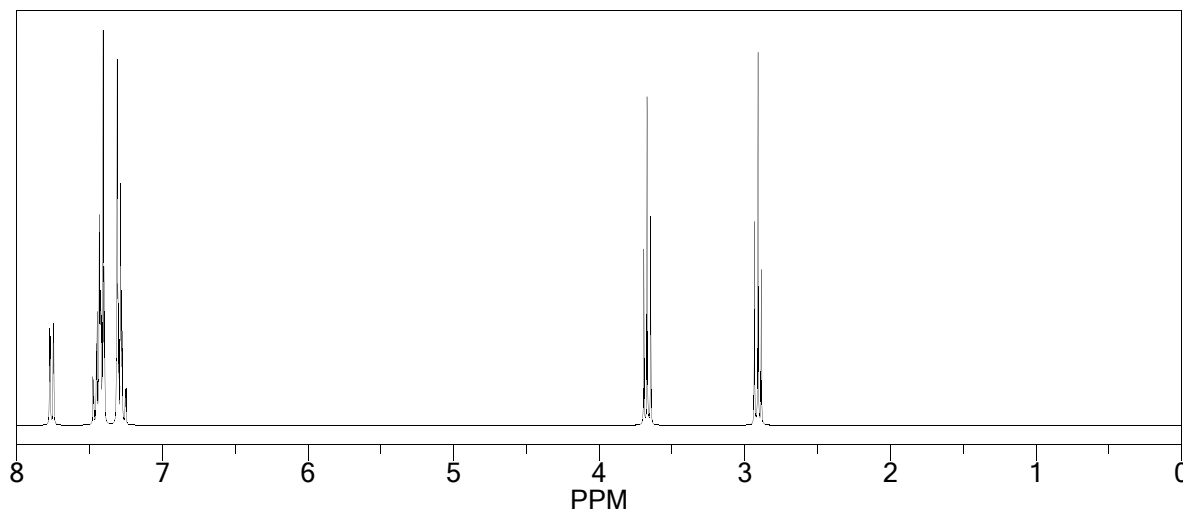
(ii) With reference to your answer in (c)(i), describe how you can easily determine the completion of reaction in step 1 **without** using any spectroscopic techniques. [1]

Step 2a requires two molar equivalents of *n*-BuLi which functions as a strong base. It abstracts protons from an N–H bond and a C–H bond in **P**.

(iii) Suggest the identities of solvent **X** and reagent **Y**. [2]

(iv) State if the mazindol obtained in this synthesis is optically pure. Explain briefly. [1]

The ^1H NMR spectrum of mazindol is shown below.



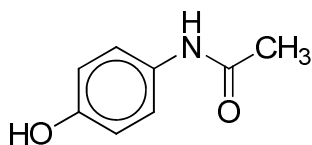
(d) (i) Assign the signals at δ (in ppm) 7.76, 3.67 and 2.91 in the ^1H NMR spectrum of mazindol. [2]

(ii) Sketch the ^1H NMR spectrum of **P** and assign all the signals. [3]

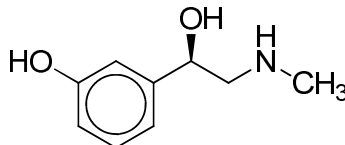
(iii) Suggest how you may be able to identify the labile proton of **P** in its ^1H NMR spectrum. [1]

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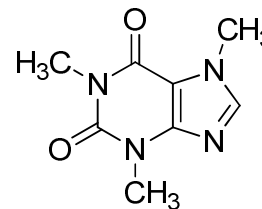
- 6 Lemsip® is a brand of cold and flu remedies widely used in many countries. One of their best known products is the Lemsip® Max Cold & Flu relief capsules which contain active ingredients paracetamol as an analgesic and phenylephrine as a decongestant.



paracetamol



phenylephrine



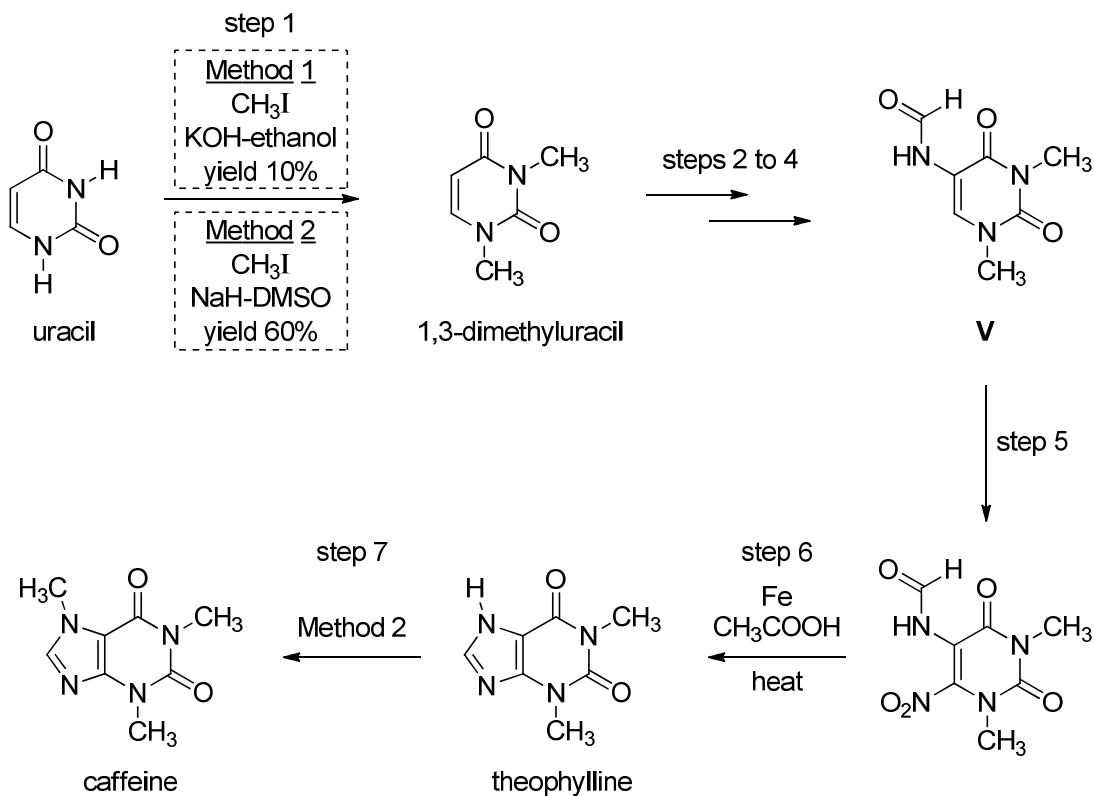
caffeine

- (a) Give **two** reasons why aspirin is **not** used as the analgesic in place of paracetamol. [2]

Lemsip® Max Cold & Flu relief capsules are available in two forms – the Day capsule and the Night capsule. Both capsules contain paracetamol and phenylephrine. However, caffeine is found only in its Day capsule and not in its Night capsule.

- (b) Suggest why caffeine is absent in the Night capsule. Give **another** physiological effect of caffeine. [2]

Although caffeine can be extracted from many natural products, it can also be synthesised from uracil in 7 steps shown below.



- (c) (i) The conversion of uracil to 1,3-dimethyluracil can be carried out via Method 1 or Method 2, with percentage yield of 10% and 60% respectively. Give **two** reasons why there is an increase in percentage yield via Method 2. [2]

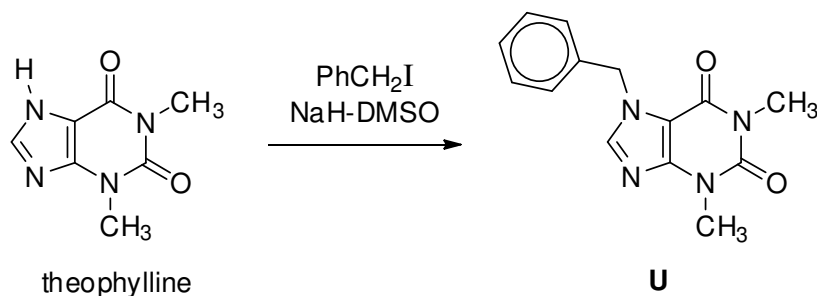
(c) (ii) For Method 2, explain whether the reaction will still proceed if NaH is replaced by LiAlH₄. [1]

(iii) By drawing a resonance structure involving the two amide bonds, show how 1,3-dimethyluracil is an aromatic compound. [1]

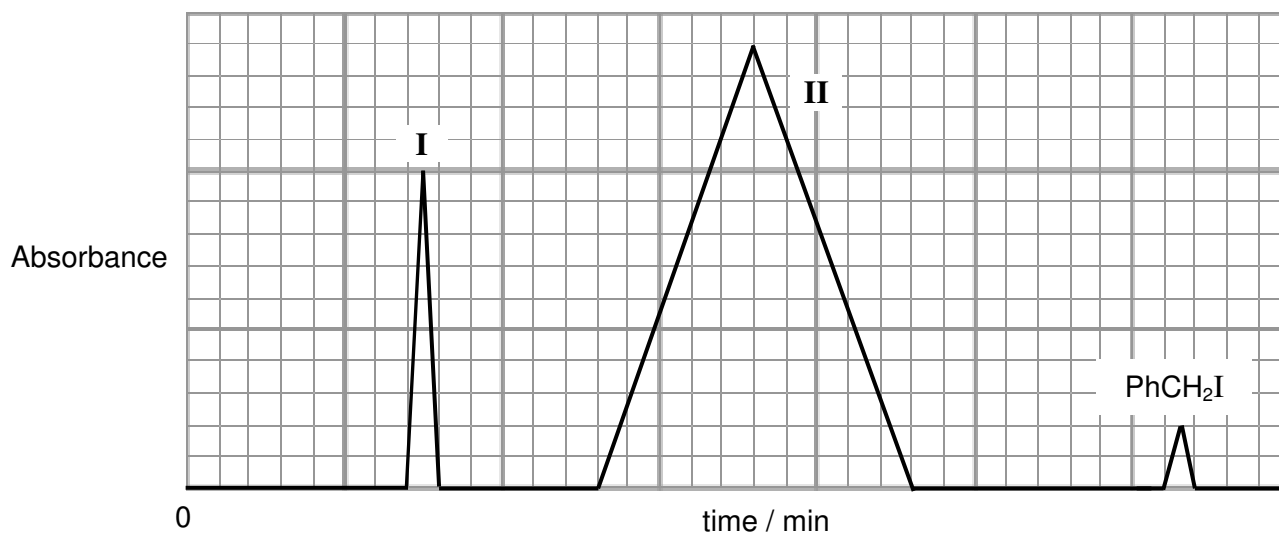
(iv) Using the information in (c)(iii), propose a 3-step synthesis of compound **V** from 1,3-dimethyluracil. [3]

(v) There are two reactions occurring in step 6. State the **two** types of reactions in step 6. [2]

To better study the effects of caffeine in the human body, a chemical biologist derivatised caffeine with hydrophobic side chains by modifying step 7 of the above scheme to obtain **U**.



After two hours, the reaction mixture was diluted with water and analysed by HPLC. The separation was carried out using an unknown stationary phase. The following chromatogram was obtained.



(d) (i) Outline the basic principles of HPLC. [2]

(ii) By considering the polarities of theophylline, PhCH₂I and **U** as well as the retention times of the three peaks, decide

- whether normal or reverse-phase HPLC was used to analyse the mixture
- which peak (**I** or **II**) corresponds to **U**

Explain your reasoning. [3]

(iii) Hence, calculate the percentage yield of **U** at this juncture. [2]

~ THE END ~